Host Interface Manual

Version 1.1

BASIC LIS RELATED TERMS

1. Control Codes

LF (Line Feed) (Hex 0A)
CR (Carriage Return) (Hex 0D)
STX (Start of Text) (Hex 02)
ETX (End of Text) (Hex 03)
EOT (End of Transmission) (Hex 04)
ENQ (Enquiry) (Hex 05)
ACK (Acknowledge) (Hex 06)
NAK (Not Acknowledged) (Hex 15)

2. Message Frame / Checksum Calculation

The end frame

[ENQ][STX] Text first char. ..... Text last char. [ETX] [EOT]

[**ENQ**]
is the ASCII code 5, indicating the request for transmission
[**STX**]
is the ASCII code 2, indicating the beginning of a frame transmission.
Text
is the data content of a frame.
[**ETX**]
is the ASCII code 3, indicating the end of the text block of an end frame.
[**EOT]**
is the ASCII code 4 indicating the end of transmission.

Example for Checksum Calculation

[ENQ][STX]1Test[ETX][EOT]

Symbol	Character	Value (decimal)
[ENQ]	[ENQ]	005
[STX]	[STX]	002
FN	‘1’	049
Text	‘T’	+084
	‘e’	+101
	‘s’	+115

	‘t’	+116
[ETX]	[ETX]	+003
[EOT]	[EOT]	+004

3. Delimiters

The Presentation Layer uses some characters as special delimiter codes. These characters may not be used as normal message text:

Delimiter	Character	Char.	ASCII decimal	ASCII hex.
Record delimiter	= carriage return	<CR>	13	0Dh
Field delimiter	= vertical bar	|	124	7Ch
Repeat delimiter	= backslash	\	92	5Ch
Component delimiter	= caret	^	94	5Eh
Escape delimiter	= ampersand	&	37	26h

4. Table of Record Message Type

Record Name	Identifier
Message Header Record	H
Patient Information Record	P
Test Order Record	O
Result Record	R
Message Terminator Record	L
Request Information Record	Q

5. Communication Options

Host communication options can be found on the System Setup/Host Communication Setting/ Communication Setting screen.
Default :9600,8,N,1.

Item	**Specifications**
Com Port	1~4
Baud rate	4800,9600,19200
data bit	7,8
Parity bit	Even, Odd, None
Stop bit	1,2

6. Retry Times Out

Enter Range : 1 - 15 (Second)

7. Data Record Usage

(refer to ASTM 1394, particularly sections 6 through 13)

Each record sent by the MAGLUMI will contain up to the last field used by the MAGLUMI, which may or may not be all fields possible for the record. The first <**Max Length**> characters are significant only. Any more characters transmitted for a specific field are ignored.

Message send from MAGLUMI to Host

Mandatory:

Y: Field is mandatory and has to be considered.

N: Field is not mandatory. The field has to be considered by the LIS, but it is transmitted only if information is available at Maglumi. When the information is not available an empty field is transmitted.

Message Header Record (H)

Description:

The Message Header Record to the host is the first record of each message.

Example:

<STX>H|\^&|||MAGLUMI|||||Host|||1|20081124151206<CR><ETX>

Field No.	E1394	ASTM Field Name	Field Contents	Max Length	Mandatory
1	7.1.1	Record Type ID	'H'	1	**Y**
2	7.1.2	Delimiter Definition	'| \ ^ &'	4	**Y**
5	7.1.5	Sender Name or ID	'MAGLUMI'	20	**Y**

10	7.1.10	Receiver ID		20	**Y**
13	7.1.13	Version No.	‘1’	1	**Y**
14	7.1.14	Date and Time of Message	‘YYYYMMDDHHMMSS’	14	**Y**

Patient Information Record (P)

Description:

This record is used to transfer patient sex and age to the host (within result message).

Example:

<STX>P|1||||ZhangDongdong|||M||||||40^Y<CR><ETX>

Field No.	E1394	ASTM Field Name	Field Contents	Max Length	Mandatory
1	8.1.1	Record Type ID	‘P’	1	**Y**
2	8.1.2	Sequence Number		6	**Y**
6	8.1.6	Patient Name		30	**N**
9	8.1.9	Patient Sex	‘M’, ‘F’ or ‘U’	1	**N**
15^1	8.1.15	Special Field 1	Age (1-200)	3	**N**
15^2			Age Unit (‘Y’,’M’,’D’)	1	

Test Order Record (O)

Description:

This record holds information regarding analysis request and sample information to the host (within result message).

Example:

<STX>O|1|CA2201320078^^1^10^N|||R|20081124151012|||||||1|||||||||O<CR><ETX>

Field No.	E1394	ASTM Field Name	Valid Contents	Max Length	Mandatory
1	9.4.1	Record Type ID	'O'	1	**Y**
2	9.4.2	Sequence Number		6	**Y**
3^1	9.4.3	Specimen ID (<Sample ID>^ <Sample No>^ <Disk ID>^ <Position No>^ <Diluent>)	Sample ID (ID Mode)	22	**Y**
3^2			Sample No. (S. No. Mode)	6	
3^3			Disk No.	1	
3^4			Position No.	2	
3^5			'Y' or 'N'	1	
6	9.4.6	Priority	'S' or 'R'	1	**Y**
7	9.4.7	Requested/Ordered Date and Time	'YYYYMMDD HHMMSS'	14	**Y**
16	9.4.16	Specimen Descriptor	1 - Serum 2 - Urine 3 - CSF 4 - Suprnt 5 - Others	1	**Y**
26	9.4.26	Report Types	'O', 'F', 'I', 'C' or 'X'	1	**Y**

Result Record (R)

Description:

The Result Record to the host contains the result of a single analytical determination.

Example:

<STX>R|1|^^^GGT|4.96|mg/dL|3^12\0^20|N||F||||20081124151012<CR><ETX>

Field No.	E 1394	ASTM Field Name	Field Contents	Max Length	Mandatory
1	10.1.1	Record Type ID	R	1	**Y**
2	10.1.2	Sequence Number		5	**Y**
3	10.1.3	Universal Test ID		10	**Y**
4	10.1.4	Data or Measurement Value		12	**N**
5	10.1.5	Units		10	**N**
6	10.1.6	Reference Ranges	first range = normal range second range = critical range separated by repeat delimiter ('\')	30	**N**
7	10.1.7	Result Abnormal Flags	'L','H','N' or 'A'	1	**N**
9	10.1.9	Result Status	'F' or 'C'	1	**Y**
13	10.1.13	Date/Time Test Completed	'YYYYMMDD HHMMSS'	14	**N**

Request Information Record to Host

Description:

The Request Information Record to the host is used to request sample program information from the host computer. A query is typically performed when a sample bar code has been read on the instrument and no sample program information is found locally. All queries are made by sample ID.

Example:

<STX>Q|1|AC2201023321^^0^1^N| |ALL|||||||O<CR><CTX>

Field No.	E1394	ASTM Field Name	Field Contents	Max Length	Mandatory
1	10.1.1	Record Type ID	'Q'	1	**Y**
2	10.1.2	Sequence Number		6	**Y**
3^1	10.1.3	Starting Range ID Number (<Sample ID>^ <Sample No>^ <Disk ID>^ <Position No>^ <Diluent>)	Sample ID (ID Mode)	22	**Y**
3^2			Sample No. (S. No. Mode)	6	
3^3			Disk No.	1	
3^4			Position No.	2	
3^5			'Y' or 'N'	1	
5	10.1.5	Universal Test ID	'ALL'		**Y**
13	10.1.13	Request Information Status Codes	'A' or 'O'	1	**Y**

Message Terminator Record

Description:

The Message Terminator Record to the host is used with the Message

Header Record to encapsulate related patient information.

Example:

<STX>L|1|N<CR><ETX>

Field No.	E1394	ASTM Field Name	Field Contents	Max Length	Mandatory
1	13.1.1	Record Type ID	'L'	1	**Y**
2	13.1.2	Sequence Number		6	**Y**
3	13.1.3	Termination Code	'N'	1	**Y**

Message send from Host to MAGLUMI

Mandatory:

Y: Field is mandatory and have to be considered.

N: Field is not mandatory. The field have to be considered by the LIS, but it is transmitted only if information is available at Maglumi. When the information is not available an empty field is transmitted.

Message Header Record (H)

Description:

The Message Header Record to the analyzer is the first record of each message.

Example:

<STX>H|\^&<CR><ETX>

Field No.	E1394	ASTM Field Name	Field Contents	Max Length	Mandatory
1	7.1.1	Record Type ID	'H'	1	**Y**
2	7.1.2	Delimiter Definition	'| \ ^ &'	4	**Y**

Patient Information Record (P)

Description:

This record is used to transfer patient sex and age to the analyzer (within

test selection message).

Example:

<STX>P|1||123456||ZhangDongdong|||M|||||40^Y<CR><ETX>

Field No.	E1394	ASTM Field Name	Field Contents	Max Length	Mandatory
1	8.1.1	Record Type ID	'P'	1	**Y**
2	8.1.2	Sequence Number		6	**Y**
4	8.1.4	Laboratory Assigned Patient ID	Sample ID or Sample No.	22	**N**
6	8.1.6	Patient Name		30	**N**
9	8.1.9	Patient Sex	'M', 'F' or 'U'	1	**N**
15^1	8.1.15	Special Field 1	Age (1-200)	3	**N**
15^2			Age Unit ('Y','M','D')	1	

Test Order Record (O)

Description:

This record holds information regarding analysis request and sample information to the analyzer (within test selection message).

Example:

<STX>O|1|AB2201022318^^1^10^N||^^^ALT|R|20081124151012|||||||| 1|||||||||O<CR><ETX>

Field No.	E1394	ASTM Field Name	Valid Contents	Max Length	Mandatory
1	9.4.1	Record Type ID	'O'	1	**Y**
2	9.4.2	Sequence Number		6	**Y**
3^1	9.4.3	Specimen ID (<Sample ID>^ <Sample No>^ <Disk ID>^ <Position No>^ <Diluent>)	Sample ID (ID Mode)	22	**Y**
3^2			Sample No. (S. No. Mode)	6	
3^3			Disk ID	1	
3^4			Position No.	2	
3^5			'Y' or 'N'	1	
5	9.4.5	Universal Test ID		10	**Y**
6	9.4.6	Priority	'S' or 'R'	1	**N**
7	9.4.7	Requested/Ordered Date and Time	'YYYYMMDD HHMMSS'	14	**N**
16	9.4.16	Specimen Descriptor	1 - Serum 2 - Urine 3 - CSF 4 - Suprnt 5 - Others	1	**N**
26	9.4.26	Report Types	'O', 'F', 'I', 'C' or 'X'	1	**N**

Message Terminator Record

Description

The Message Terminator Record to the analyzer is used with the Message Header Record to encapsulate related patient information.

Example:

<STX>L|1|N<CR><ETX>

Field No.	E1394	ASTM Field Name	Field Contents	Max Length	Mandatory
1	13.1.1	Record Type ID	'L'	1	**Y**
2	13.1.2	Sequence Number		6	**Y**
3	13.1.3	Termination Code	'N' or 'I'	1	**Y**

notice：
When the double direction communication apply the testing from the MAGLUMI To Lis system，If when the lis system didn'f find the sample that should be applied,must be send:
ENQ
EOT